

Econ 30065: Microeconomics

Sara Negrelli

Oligopoly (II)

Cournot quantity competition

- **Cournot model of oligopoly:** firms choose how much to produce (Q) simultaneously, and the price clears the market given the total quantity produced
- Assume homogeneous goods
- To keep things simple we study a duopoly

How to solve the game?

- Simultaneous moves game
- Infinite set of actions
- *Thus:*
 - 1- find best response functions
 - 2- see where they cross each other: that is NE

Firms' profits

Firm 1:

$$P \times Q_1 - TC_1$$

Firm 2:

$$P \times Q_2 - TC_2$$

Where P is the (total) inverse demand: $P(Q_1 + Q_2)$

Step 1: Best responses

- Take derivative of profits for each firm
 - ➔ Set MR equal to MC for each firm
- $MR_1 = MC_1$
- $MR_2 = MC_2$
- ➔ find best response functions!

Step 2: Nash Equilibrium

Solve the system of equations:

$$Q_1 = f(Q_2)$$

$$Q_2 = f(Q_1)$$

➔ Find NE: (Q_1, Q_2)

➔ To find P use inverse demand function

Example

Assume two firms are competing on quantities. They choose simultaneously a quantity of the good they want to sell. Their total costs are given by $40Q$. The market demand is given by $Q^D = 10000 - 100P$.

What is the Nash equilibrium in the game?

Example

Profits:

Firm 1:

$$\text{Profits} = P \times Q_1 - 40Q_1$$

$$\text{where } P(Q) = 100 - 0.01(Q_1 + Q_2)$$

$$\rightarrow \text{Profits} = (100 - 0.01(Q_1 + Q_2)) \times Q_1 - 40Q_1$$

Firm 2:

$$\text{Profits} = P \times Q_2 - 40Q_2$$

$$\text{where } P(Q) = 100 - 0.01(Q_1 + Q_2)$$

$$\rightarrow \text{Profits} = (100 - 0.01(Q_1 + Q_2)) \times Q_2 - 40Q_2$$

Example: best response functions

$$TR_1 = (100 - 0.01Q_1 - 0.01Q_2)Q_1$$

$$MR_1 = 100 - 0.02Q_1 - 0.01Q_2$$

$$MR_1 = MC: 100 - 0.02Q_1 - 0.01Q_2 = 40$$

→ $Q_1 = 3000 - 0.5Q_2$ best response function for firm 1

$$TR_2 = (100 - 0.01Q_2 - 0.01Q_1)Q_2$$

$$MR_2 = 100 - 0.02Q_2 - 0.01Q_1$$

$$MR_2 = MC: 100 - 0.02Q_2 - 0.01Q_1 = 40$$

→ $Q_2 = 3000 - 0.5Q_1$ best response function for firm 2

Example: find NE

$$Q_1 = 3000 - 0.5Q_2$$

$$Q_2 = 3000 - 0.5Q_1$$

$$Q_1 = 3000 - 0.5(3000 - 0.5Q_1)$$

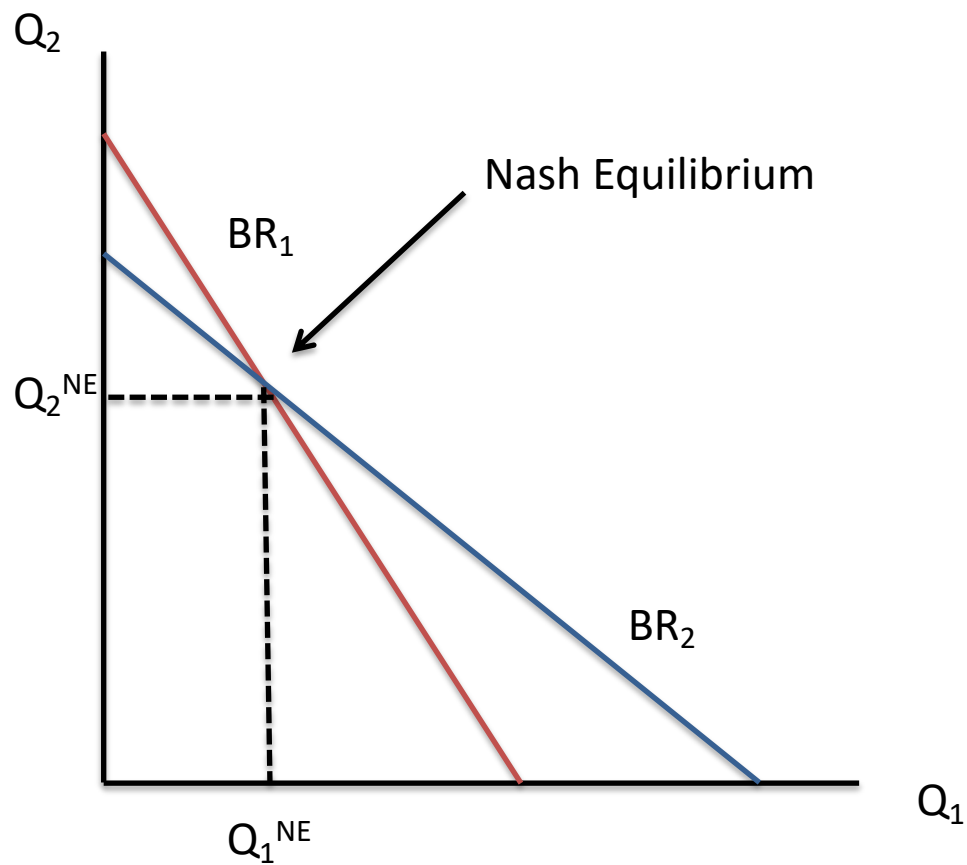
$$Q_1 = 2000$$

$$Q_2 = 2000$$

To find the price: use inverse demand function:

$$P = 100 - 0.01(2000 + 2000) = 60$$

Graphically



Best response functions with identical firms

When two firms have the same cost structure, and face the same demand, they have the same best response function and thus they will produce the same amount of output.

$$Q_1 = 3000 - 0.5Q_2$$

$$Q_2 = 3000 - 0.5Q_1$$

Cournot competition vs Bertrand Model

- Important difference from equilibrium in Bertrand market: in Cournot, equilibrium price is always above marginal cost
- Deadweight loss in Cournot!

Deadweight Loss from Duopoly versus Monopoly

- Will the aggregate surplus be higher in the Bertrand and Cournot equilibrium compared to the monopoly outcome?
- In Bertrand Model: $DWL=0$
- In Cournot Model: DWL smaller than in Monopoly
 - The deadweight loss of monopoly is larger because the monopoly price is further above marginal cost than is the oligopoly price